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Scuola di Ingegneria Industriale e dell'Informazione
Corso di Laurea Magistrale in Mathematical Engineering – Quantitative Finance

Computational Finance: Energy Project

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1 Dataset description and statistical analysis

The dataset `power_DE_FR` contains daily prices of German (DE) and French (FR) electricity baseload swap contracts over the period from 02 January 2015 to 04 November 2025. Contracts are classified by delivery structure into monthly (Mc1–Mc4), quarterly (Qc1–Qc4) and yearly (Yc1–Yc3) products. The original dataset included 22 contracts. Following a preliminary screening, the two longest-dated yearly contracts (`TRDEBYc3` and `TRFRBYc3`) were removed, resulting in 20 time series (10 DE and 10 FR). A row-filtering procedure was applied to eliminate extended blocks of missing quotations, leading to the exclusion of 457 observations (16.4% of the sample). The final cleaned dataset consists of 2326 daily observations.

1.2 Statistical properties of log-returns

Sample means of daily log-returns are close to zero across all maturities, confirming the absence of systematic trends at the daily frequency. Return volatility ranges from 1.9% to 5.8%, with short-dated contracts exhibiting higher volatility, consistently with the Samuelson effect typically observed in electricity markets. Higher-order moments reveal pronounced departures from Gaussianity: skewness ranges from -6.13 to 1.93, kurtosis is high (12.6-125.4), indicating asymmetric and heavy-tailed distributions. Overall, while log-returns are clearly non-Gaussian, their first and second moments are well defined and stable. This supports the use of a Gaussian market model as a pragmatic approximation, consistently with the approach adopted by Gardini and Santilli (2023).

1.3 Stationarity, normality and independence

ADF tests reject the presence of a unit root for all series at standard significance levels (p-values $\approx 10^{-3}$), indicating that daily log-returns are stationary. In energy markets log-returns are usually not normally distributed, their distribution is often skewed and presents heavy tails or fat tails effect. Jarque–Bera tests reject normality for all contracts, confirming the presence of skewness and heavy tails. In light of this results, by choosing a normal distribution we might lose some market peculiarities. On the other hand, in order to get a simple and stable calibration methodology the hypothesis of normality in log-returns is commonly accepted, following the approach adopted in the paper of Gardini and Santilli.
Levy processes has independent increments. For this reason, before using Levy processes for modeling purposes, we have to check that the increments are independent. Sample autocorrelation analysis shows no significant serial dependence beyond lag zero, supporting the independent-increments assumption.

1.4 Cross-sectional dependence and modeling implications

Cross-sectional dependence is strong. Pairwise daily correlations range from -0.15 to 0.88. Average within-market correlations equal 0.59 for German contracts and 0.44 for French contracts, while the average cross-market correlation is 0.49, indicating a substantial degree of market integration. Overall, the strong and persistent dependence structure suggests that the joint dynamics of German and French electricity swaps can be effectively captured by a small number of common risk factors. This provides a clear motivation for the use of a reduced-factor HJM market model calibrated via Principal Component Analysis.

2 Fit of the Monthly Forward Curve

To construct the monthly forward curve, we initially formulated a linear system of the form

$$Ax = b,$$

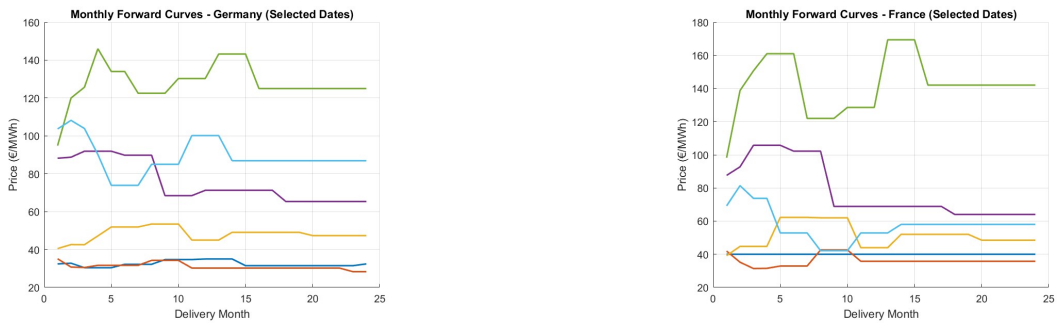
where b collects the observed market prices available in the dataset, x represents the vector of unknown implicit monthly forward prices, and A is a weighting matrix whose rows sum to one and whose entries encode the fraction of each observed contract price attributable to the corresponding monthly deliveries.

However, even after removing all equations (i.e. rows of the system) for which the vector of observed prices contained missing values, the resulting linear system turned out to be heavily underdetermined. This issue stems from the limited overlap between contracts with different delivery granularities, which prevents the system from providing sufficient independent constraints to uniquely identify the monthly prices.

For this reason, we adopted a more pragmatic, brute-force construction procedure. For each observation date, we proceeded hierarchically by first fixing, whenever available, the prices of monthly contracts. We then incorporated quarterly contracts, allocating their values across the corresponding monthly entries that were still unassigned, in such a way as to preserve both the original quarterly average and the monthly values previously fixed. The same procedure was subsequently applied to yearly contracts. Finally, any remaining unfilled monthly entries were set equal to the global mean price.

This approach is extremely robust from a numerical standpoint. However, the structure of the provided dataset does not appear to be well suited for a fine monthly decomposition. In particular, the very limited number of overlapping contracts across different delivery horizons implies that many columns of the resulting monthly grid are effectively identical, as they are obtained by uniformly splitting the same price associated with a coarser delivery period. As a consequence, the fitted monthly forward curve contains a substantial degree of redundancy inherited from the original data structure.

(ii) Statistical analysis of monthly forward curves

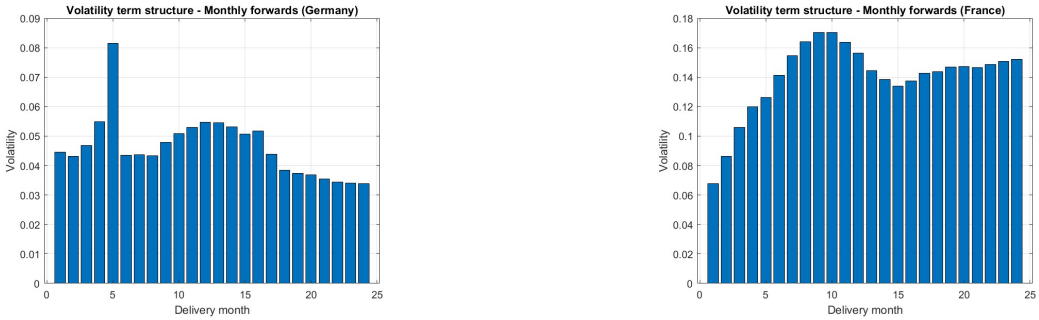


(a) Germany (b) France

Figure 1: Reconstructed monthly forward curves for selected observation dates (delivery months $M1$–$M24$).

Sample means of monthly forward log-returns are close to zero for both markets. Return volatility is heterogeneous across maturities: the standard deviation ranges from 3.4% to 8.2% for German forwards and from 6.8% to 17.1% for French forwards.

Higher-order moments indicate pronounced deviations from Gaussianity. Skewness for German forwards skewness ranges from -5.16 to 0.54 and kurtosis from 11.4 to 484.0, while French forwards exhibit milder asymmetry (skewness between -0.14 and 1.17) and kurtosis between 8.7 and 32.6.



(a) Germany (b) France

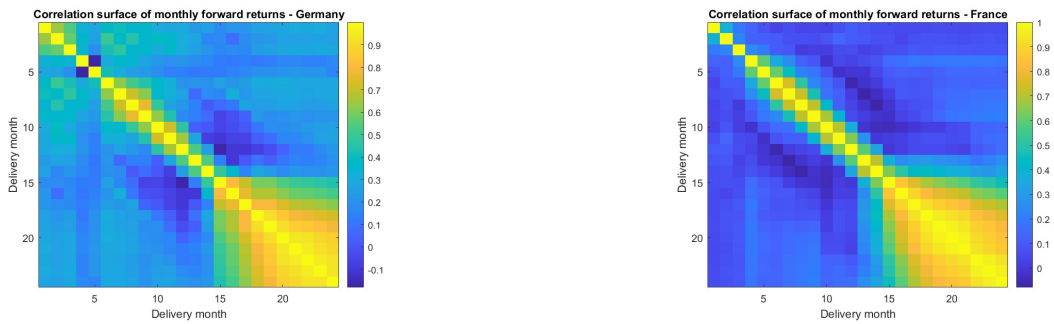
Figure 2: Volatility term structure of monthly forward log-returns (Samuelson effect).

Stationarity and dependence structure

Augmented Dickey–Fuller tests reject the unit-root hypothesis for all 24 monthly forwards in both markets, while Jarque–Bera tests reject normality for all series. The Samuelson effect is clearly visible for German forwards, with front-month volatility exceeding long-maturity volatility by a factor of about 1.31.

Cross-sectional dependence across maturities is strong: average pairwise correlations equal 0.32 for German forwards and 0.25 for French forwards. Cross-market correlations between German and French forwards of the same maturity are positive but moderate.

Overall, the statistical properties of reconstructed monthly forwards closely mirror those of the underlying swap contracts and support the use of a reduced-factor HJM market model calibrated via Principal Component Analysis.



(a) Germany (b) France

Figure 3: Correlation surface of monthly forward log-returns, showing strong cross-maturity dependence.

3 PCA on Forward Returns

As a first preprocessing step, we select in the original dataset all observations containing an insufficient amount of information to be considered reliable. In particular, the rows exhibiting two or more missing values in the monthly observations, two or more missing values in the quarterly observations, or at least one missing value in the yearly observations.

Then we apply Principal Component Analysis (PCA) to the forward *log-returns* of Germany and France forwards in order to reduce the number of risk factors.

Returns construction and cleaning Daily log-returns are computed as

$$R_t = \log(F_{t+1}) - \log(F_t).$$

We keep a return only if both consecutive prices are valid (not flagged by the first preprocessing step). In addition, we keep only *intra-month* returns (two consecutive dates must belong to the same calendar month) to avoid artificial jumps due to rolling between months.

Outlier removal Before PCA, we remove multivariate outliers using the squared Mahalanobis distance

$$d_t^2 = (R_t - \mu)^\top S^\dagger (R_t - \mu),$$

and we discard observations with $d_t^2 > \chi_r^2(0.995)$, where $r = \mathrm{rank}(S)$. This filtering step improves the numerical stability of the covariance estimate and, in turn, the robustness of the PCA results.

The Mahalanobis-based criterion relies on the assumption of joint normality of the return vector. In practice, this assumption is violated, as energy returns are known to exhibit heavy tails and deviations from Gaussian behavior. As a consequence, the chi-square threshold does not control the nominal false positive rate, leading to the identification of a substantially larger number of observations than theoretically expected under joint normality. In our dataset, the procedure flags 287 observations out of 1936 as outliers.

PCA and variance threshold PCA is applied to the cleaned return matrix (centered). We select the smallest number of components m such that the cumulative explained variance satisfies

$$\sum_{k=1}^{m} \mathrm{Explained}_k \geq 90\%.$$

We choose **90%** as a reasonable trade-off: it captures most of the systematic co-movements of the forward curve while avoiding additional components that typically explain only small residual variance (often closer to noise), and it keeps the factor dimension low for calibration and simulation.

4 Chooser Option Pricing

We consider a chooser option with payoff

$$\max(F_T^{(1)}, F_T^{(2)}),$$

where $F_T^{(1)}$ and $F_T^{(2)}$ denote respectively the German and French monthly forward prices observed at maturity T.

The time-t_0 price is

$$\mathrm{Price} = D(t_0, T)\, \mathbb{E}\Big[\max(F_T^{(1)}, F_T^{(2)})\Big].$$

4.1 Derivation of the Closed-Form Formula

We start by decomposing the expectation:

$$\mathbb{E}[\max(F_T^{(1)}, F_T^{(2)})] = \mathbb{E}[F_T^{(1)}\mathbf{1}_{\{F_T^{(1)}>F_T^{(2)}\}}] + \mathbb{E}[F_T^{(2)}\mathbf{1}_{\{F_T^{(2)}>F_T^{(1)}\}}] =$$

$$= \mathbb{E}[F_T^{(1)} \mid F_T^{(1)} > F_T^{(2)}]\mathbb{P}(F_T^{(1)} > F_T^{(2)}) + \mathbb{E}[F_T^{(2)} \mid F_T^{(2)} > F_T^{(1)}]\mathbb{P}(F_T^{(2)} > F_T^{(1)}).$$

In order to compute these quantities, it is convenient to rewrite the events in terms of a ratio. Indeed,

$$\mathbb{P}(F_T^{(1)} > F_T^{(2)}) = \mathbb{P}\left(\frac{F_T^{(1)}}{F_T^{(2)}} > 1\right), \qquad \mathbb{P}(F_T^{(2)} > F_T^{(1)}) = \mathbb{P}\left(\frac{F_T^{(2)}}{F_T^{(1)}} > 1\right).$$

This allows us to study a single random variable, $X_T := \frac{F_T^{(2)}}{F_T^{(1)}}$.
We assume that the forward prices follow

$$\frac{dF_t^{(i)}}{F_t^{(i)}} = \sigma_i \cdot dW_t, \qquad i = 1, 2, \tag{1}$$

where $W_t \in \mathbb{R}^N$ is a Brownian motion with independent components and $\sigma_i \in \mathbb{R}^N$ are volatility vectors.

Construction of σ_1 and σ_2. Following the reference paper, the N-factor loading matrix is defined as $\sigma^* = C^*(\Gamma^*)^{1/2}\Delta t^{-1/2}, \Delta t = 1/252$, where C^* contains the first N eigenvectors of the covariance matrix of daily log-returns and Γ^* is the diagonal matrix of the corresponding eigenvalues. In particular, the vectors σ_1 and σ_2 used in (1) are the rows of σ^* corresponding to the German and French monthly forwards with delivery starting on 5 April 2027:

$$\sigma_1 = \sigma^*_{\text{DE, Apr 2027}}, \qquad \sigma_2 = \sigma^*_{\text{FR, Apr 2027}}.$$

The solution of (1) is

$$F_t^{(i)} = F_0^{(i)} \exp\left(-\tfrac{1}{2}\|\sigma_i\|^2 t + \sigma_i \cdot W_t\right),$$

Defining $X_t = F_t^{(2)}/F_t^{(1)}$ and applying Itô's formula, one obtains

$$dX_t = X_t\big(\|\sigma_1\|^2 - \sigma_1 \cdot \sigma_2\big)dt + X_t(\sigma_2 - \sigma_1) \cdot dW_t,$$

which shows that X_t follows a geometric Brownian motion. Its explicit solution is

$$X_t = X_0 \exp\left(\tfrac{1}{2}(\|\sigma_1\|^2 - \|\sigma_2\|^2)t + (\sigma_2 - \sigma_1) \cdot W_t\right).$$

Since $W_t \sim \mathcal{N}(0, tI)$, the scalar product $(\sigma_2 - \sigma_1) \cdot W_t$ is normally distributed with variance $\|\sigma_2 - \sigma_1\|^2 t$. Hence,

$$\ln X_t \sim \mathcal{N}\left(\ln X_0 + \tfrac{1}{2}(\|\sigma_1\|^2 - \|\sigma_2\|^2)t,\ \|\sigma_2 - \sigma_1\|^2 t\right),$$

so X_t is lognormally distributed. In particular,

$$\mathbb{P}(F_T^{(2)} > F_T^{(1)}) = \mathbb{P}(X_T > 1) = \Phi\left(\frac{\ln(F_0^{(2)}/F_0^{(1)}) + \frac{1}{2}(\|\sigma_1\|^2 - \|\sigma_2\|^2)T}{\|\sigma_2 - \sigma_1\|\sqrt{T}}\right).$$

Similarly,

$$\mathbb{P}(F_T^{(1)} > F_T^{(2)}) = \Phi\left(\frac{\ln\big(F_0^{(1)}/F_0^{(2)}\big) + \frac{1}{2}\big(\|\sigma_2\|^2 - \|\sigma_1\|^2\big)T}{\|\sigma_1 - \sigma_2\|\sqrt{T}}\right) = 1 - \mathbb{P}(F_T^{(2)} > F_T^{(1)}).$$

Moreover, from

$$\ln F_T^{(i)} = \ln F_0^{(i)} - \tfrac{1}{2}\|\sigma_i\|^2 T + \sigma_i \cdot W_T,$$

Since $W_T \sim \mathcal{N}(0, TI)$ is a Gaussian vector and each $\ln F_T^{(i)}$ is an affine (linear + constant) function of W_T, the pair

$$(\ln F_T^{(1)}, \ln F_T^{(2)}) = A\,W_T + b$$

for suitable $A \in \mathbb{R}^{2\times n}$ and $b \in \mathbb{R}^2$ is a linear transformation of a Gaussian vector. Therefore, $(\ln F_T^{(1)}, \ln F_T^{(2)})$ is jointly Gaussian (bivariate normal). In particular,

$$\mathrm{Var}(\ln F_T^{(i)}) = \|\sigma_i\|^2 T, \qquad \mathrm{Cov}(\ln F_T^{(1)}, \ln F_T^{(2)}) = (\sigma_1 \cdot \sigma_2)T,$$

so that the correlation of the log-prices is

$$\rho = \frac{\mathrm{Cov}(\ln F_T^{(1)}, \ln F_T^{(2)})}{\sqrt{\mathrm{Var}(\ln F_T^{(1)})\mathrm{Var}(\ln F_T^{(2)})}} = \frac{\sigma_1 \cdot \sigma_2}{\|\sigma_1\|\,\|\sigma_2\|}.$$

Denote

$$\ln F_T^{(i)} \sim \mathcal{N}(m_i, s_i^2), \qquad m_i = \ln F_0^{(i)} - \tfrac{1}{2}\|\sigma_i\|^2 T, \qquad s_i = \|\sigma_i\|\sqrt{T}.$$

Thanks to the bivariate normal structure, the conditional expectation appearing in the payoff decomposition admits a closed form. In particular,

$$\mathbb{E}\Big[F_T^{(1)} \mid F_T^{(1)} > F_T^{(2)}\Big] = e^{m_1 + \frac{1}{2}s_1^2} \frac{\Phi\left(\frac{m_1 - m_2 + s_1^2 - \rho s_1 s_2}{\sqrt{s_1^2 + s_2^2 - 2\rho s_1 s_2}}\right)}{\Phi\left(\frac{m_1 - m_2}{\sqrt{s_1^2 + s_2^2 - 2\rho s_1 s_2}}\right)},$$

and symmetrically,

$$\mathbb{E}\Big[F_T^{(2)} \mid F_T^{(2)} > F_T^{(1)}\Big] = e^{m_2 + \frac{1}{2}s_2^2} \frac{\Phi\left(\frac{m_2 - m_1 + s_2^2 - \rho s_1 s_2}{\sqrt{s_1^2 + s_2^2 - 2\rho s_1 s_2}}\right)}{\Phi\left(\frac{m_2 - m_1}{\sqrt{s_1^2 + s_2^2 - 2\rho s_1 s_2}}\right)}.$$

Combining all terms, the final closed-form price is obtained by plugging these expressions into the initial decomposition.

$$\boxed{\text{Price} = D(t_0, T)\left(\mathbb{E}[F_T^{(1)} \mid F_T^{(1)} > F_T^{(2)}]\mathbb{P}(F_T^{(1)} > F_T^{(2)}) + \mathbb{E}[F_T^{(2)} \mid F_T^{(2)} > F_T^{(1)}]\mathbb{P}(F_T^{(2)} > F_T^{(1)})\right).}$$

This is a fully explicit closed-form expression, based only on normal cumulative distribution functions. The numerical result is reported in Table 1 below.

4.2 Verification via Margrabe Formula

Using the identity

$$\max\big(F_T^{(1)}, F_T^{(2)}\big) = F_T^{(2)} + \big(F_T^{(1)} - F_T^{(2)}\big)^+,$$

we only need to price the exchange option $\big(F_T^{(1)} - F_T^{(2)}\big)^+$. Under the standard lognormal setting (two tradables driven by the same Brownian motion vector), the Margrabe formula gives the *time-t_0 value* of the exchange option as

$$V_{\text{ex}}(t_0) = F_{t_0}^{(1)}\Phi(d_1) - F_{t_0}^{(2)}\Phi(d_2),$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function and

$$d_1 = \frac{\ln\big(F_{t_0}^{(1)}/F_{t_0}^{(2)}\big) + \frac{1}{2}\,\|\sigma_1 - \sigma_2\|^2\,(T - t_0)}{\|\sigma_1 - \sigma_2\|\,\sqrt{T - t_0}},$$

$$d_2 = d_1 - \|\sigma_1 - \sigma_2\|\,\sqrt{T - t_0}.$$

Here $\sigma_1, \sigma_2 \in \mathbb{R}^m$ are the (constant) volatility loading vectors of $\ln F^{(1)}$ and $\ln F^{(2)}$ under the same measure, hence the effective exchange-option volatility is $\sigma_{\text{ex}} = \|\sigma_1 - \sigma_2\|$.

Therefore, the Margrabe benchmark for the original payoff is

$$V_{\max}(t_0) = F_{t_0}^{(2)} + V_{\text{ex}}(t_0) = F_{t_0}^{(2)} + F_{t_0}^{(1)}\Phi(d_1) - F_{t_0}^{(2)}\Phi(d_2).$$

This value is used as a benchmark to validate the closed-form implementation and the Monte Carlo estimator.

4.3 Upper and Lower Bounds

Since

$$\max(F_T^{(1)}, F_T^{(2)}) \le F_T^{(1)} + F_T^{(2)},$$

the upper bound follows as

$$\text{Price} \le D(t_0, T)\left(\mathbb{E}[F_T^{(1)}] + \mathbb{E}[F_T^{(2)}]\right).$$

Similarly,

$$\mathbb{E}[\max(F_T^{(1)}, F_T^{(2)})] \ge \max(\mathbb{E}[F_T^{(1)}], \mathbb{E}[F_T^{(2)}]),$$

giving the lower bound

$$\text{Price} \ge D(t_0, T)\max(\mathbb{E}[F_T^{(1)}], \mathbb{E}[F_T^{(2)}]).$$

4.4 Monte Carlo Pricing

As a further validation, we simulate the dynamics of $F_T^{(1)}, F_T^{(2)}$ using

$$F_T^{(i)} = F_0^{(i)}\exp\Big(-\tfrac{1}{2}\|\sigma_i\|^2 T + \sqrt{T}\,\sigma_i Z\Big),$$

where $Z \sim \mathcal{N}(0, I_n)$. The price is estimated by

$$\text{Price}_{\text{MC}} = D(t_0, T)\,\frac{1}{N}\sum_{k=1}^{N}\max(F_T^{(1,k)}, F_T^{(2,k)}).$$

Monte Carlo is not required for pricing in this case, since a closed-form solution exists, but it is useful for numerical verification and robustness checks.

4.5 Numerical results

We now report the numerical price estimates obtained with the three pricing methods, together with lower and upper bounds.

Method	**Value**
Closed-form	98.43
Monte Carlo	98.32
Monte Carlo (95% CI)	[97.85 , 98.78]
Margrabe	98.43
Lower bound	81.72
Upper bound	136.45

Table 1: Chooser option price estimates and bounds, discounted to t_0. The Monte Carlo estimate is computed with $N = 100{,}000$ simulations. The reported confidence interval corresponds to a 95% confidence level.

5 Daily Swing Option Pricing

We price a *daily swing call* on German electricity with *upswings only* over the delivery window

$$[t_s, t_e] = \text{01-Nov-2027 to 30-Nov-2027}.$$

Exercise is allowed only on business days: removing weekends leaves $n = 22$ exercise dates in November 2027.

5.1 Contract definition

Let $\{S_{t_j}\}_{j=1}^{n}$ be the daily underlying price on the business-day grid of the delivery period. The holder can exercise at most N times ("rights") over the period. If exercised at date t_j, the payoff is

$$\Pi(t_j) = Q\,(S_{t_j} - K)^+, \qquad Q = 1.$$

We study how the option value changes with N (maximum number of daily exercises) and with the strike K.

5.2 Numerical approach: recombining binomial tree + dynamic programming

Given the short horizon (one month) and the integer control variable (remaining rights), we implement a discrete-time dynamic program on a *recombining* binomial tree.

Underlying tree (daily step). On the daily step $\Delta t = 1/252$, we build a CRR-type recombining tree:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}},$$

and at node (i, j) (state i, time index j) the underlying is

$$S_{i,j} = S_0\, u^{j-i}\, d^{i-1}, \qquad 1 \le i \le j \le n.$$

Discounting on the exercise grid. Let $DF(t_0, t)$ be the discount factor from pricing date t_0 to date t. Interpolating DF on $\{t_j\}$, we compute the one-step discount factors

$$\delta_j = \frac{DF(t_0, t_{j+1})}{DF(t_0, t_j)}, \qquad j = 1, \ldots, n-1,$$

so that one-day continuation values are multiplied by δ_j.

Recursion with remaining rights. Let $V_{i,j}^{(z)}$ be the option value at node (i, j) when $z \in \{1, \ldots, N\}$ rights remain. At the last exercise date (terminal layer),

$$V_{i,n}^{(z)} = (S_{i,n} - K)^+, \qquad z = 1, \ldots, N.$$

For $j = n-1, \ldots, 1$ we compare:

- **Continue** (do not exercise at t_j):

$$C_{i,j}^{(z)} = \delta_j \, \frac{V_{i,j+1}^{(z)} + V_{i+1,j+1}^{(z)}}{2}.$$

- **Exercise** (use one right at t_j):

$$E_{i,j}^{(z)} = (S_{i,j} - K)^+ + \delta_j \, \frac{V_{i,j+1}^{(z-1)} + V_{i+1,j+1}^{(z-1)}}{2}, \qquad z \geq 2.$$

  For $z = 1$, this reduces to the standard American-call recursion:

$$V_{i,j}^{(1)} = \max\Big\{(S_{i,j} - K)^+,\, C_{i,j}^{(1)}\Big\}.$$

Hence, for $z \geq 2$:

$$V_{i,j}^{(z)} = \max\Big\{C_{i,j}^{(z)},\, E_{i,j}^{(z)}\Big\}.$$

The swing price at t_0 is obtained from the root node and discounted to t_0:

$$\text{SwingPrice}(N) = V_{1,1}^{(N)} \, DF(t_0, t_1).$$

Constructive valuation for a fixed N. For a fixed exercise cap N, we solve the problem in N steps. First we price the case with one right left ($z = 1$), which is just a standard American call on the same tree. Then we price $z = 2$, and so on up to $z = N$. This is *constructive* because the step for z needs the values from $z - 1$: if we exercise today, we add the payoff and continue with one fewer right.

5.3 Results, checks, and dependence on N

Monotonicity and strip upper bound. The value is non-decreasing in the number of rights:

$$\text{Price}(N+1) \geq \text{Price}(N).$$

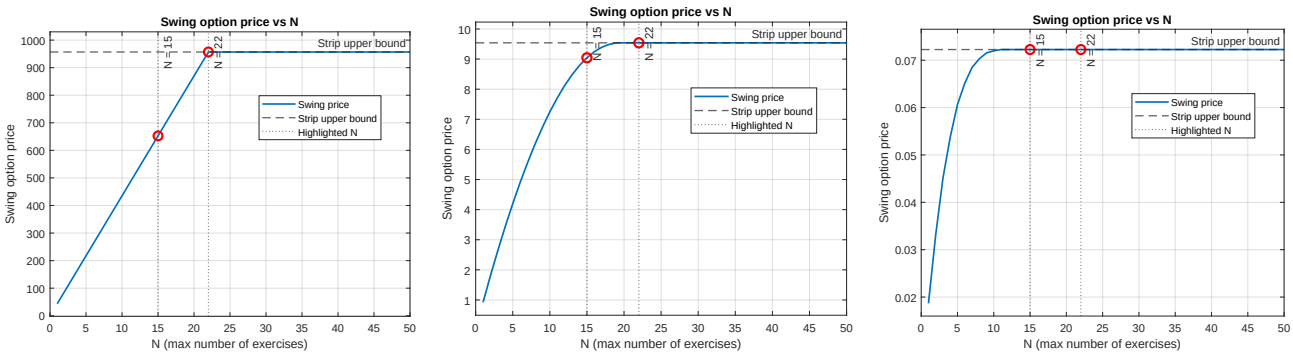
A natural *upper bound* is the value of exercising independently every day, i.e. a *strip* of daily European calls (one per business day in the delivery month). When $N \geq n = 22$, the swing constraint is non-binding and the swing value should coincide with this strip.

Pricing results
Swing price ($N = 15$): **652.5433**
Swing price ($N = 22$): **956.9284**
Sum of daily European calls (strip): **956.9284**

Table 2: Swing option prices for selected values of N and comparison with the daily-call strip.

Sanity check vs. strip. For the strike $K = 40$ (deep ITM) we report the required case $N = 15$ and the non-binding case $N = 22$ (all business days). As expected, for $N = 22$ the swing price matches the strip value.

Price vs. N and moneyness (varying K).
Figure 4 shows Price(N) together with the strip upper bound for three strikes. Across all cases, **Price**(N) increases with N and **saturates** to the strip value as $N \to 22$ (constraint becomes irrelevant). The **level** of the curve is **strictly decreasing in** K (call-style payoff). The **speed of saturation** depends on moneyness: **Deep ITM ($K = 40$):** exercising is optimal on most days, so additional rights have an almost constant marginal value until the strip cap is reached (nearly linear region). **OTM / deep OTM ($K = 90, 100$):** profitable days are rarer, so the curve is steeper for small N and becomes quickly concave, approaching the strip bound earlier.



$K = 40$ (deep ITM) $K = 90$ (OTM) $K = 100$ (deep OTM)

Figure 4: Swing option price vs. N for different strikes. The dashed line is the strip upper bound (sum of daily European calls).

5.4 Takeaways

The swing option is most valuable when the exercise limit is *binding*, i.e. when N is small and the holder must choose carefully on which days to exercise. In out-of-the-money scenarios this flexibility is even more relevant, because profitable exercise opportunities are sporadic and timing matters. As N increases and approaches the number of available business days ($n = 22$), the constraint becomes effectively non-binding: the holder can exercise on every day that is optimal. In this regime, the swing payoff is equivalent to a *strip* of daily European calls (one maturity per business day), so the swing price converges to the strip value. This provides a convenient sanity check for the implementation.